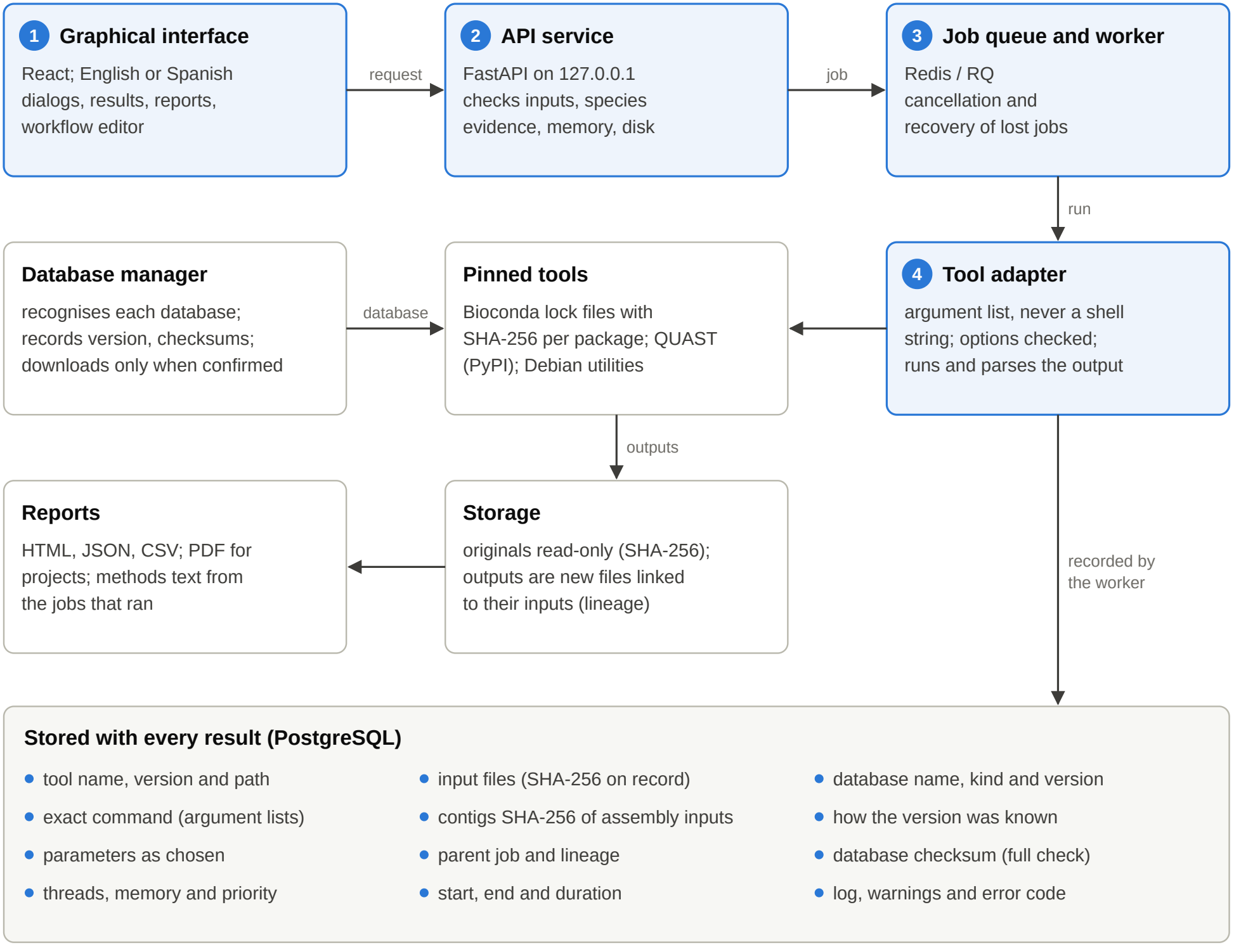
1 Graphical interface
React; English or Spanish
dialogs, results, reports,
workflow editor
request
2 API service
FastAPI on 127.0.0.1
checks inputs, species
evidence, memory, disk
job
3 Job queue and worker
Redis / RQ
cancellation and
recovery of lost jobs
run
Database manager
recognises each database;
records version, checksums;
downloads only when confirmed
database
Pinned tools
Bioconda lock files with
SHA-256 per package; QUAST
(PyPI); Debian utilities
4 Tool adapter
argument list, never a shell
string; options checked;
runs and parses the output
outputs
Reports
HTML, JSON, CSV; PDF for
projects; methods text from
the jobs that ran
Storage
originals read-only (SHA-256);
outputs are new files linked
to their inputs (lineage)
recorded by
the worker
Stored with every result (PostgreSQL)
tool name, version and path
exact command (argument lists)
parameters as chosen
threads, memory and priority
input files (SHA-256 on record)
contigs SHA-256 of assembly inputs
parent job and lineage
start, end and duration
database name, kind and version
how the version was known
database checksum (full check)
log, warnings and error code